

# **Essays in Empirical Finance**

A thesis submitted to The University of Manchester  
for the degree of PhD  
in the Faculty of Humanities.

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# List of publications

Reader, Avery. & Co-author, A. (2026). *Title of the first paper*. Journal Name, in press.

Reader, Avery., Co-author, A. & Co-author, B. (2026). *Title of the second paper*. Manuscript under review.

Reader, Avery. & Co-author, C. (2026). *Title of the third paper*. Manuscript in preparation.

# Abstract

The University of Manchester

Avery L Reader

PhD

*Essays in Empirical Finance*

2027

Write your abstract here. The University policy (section 8.1.e) requires that the abstract not exceed one page; aim for under 350 words at 12pt with 1.5 line spacing. State the research question, the methodological approach, the principal findings, and the contribution.

# **Declaration of originality**

I declare that no portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.



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# Acknowledgements

Write your acknowledgements here. Policy section 9.1.b allows a dedication and acknowledgements after the compulsory preliminary pages. Acknowledgements do not count toward the main-text word limit.

# Chapter 1

## Introduction

### 1.1 Research objectives and motivation

State the motivation for the research and the overarching question(s) the thesis addresses. The field has evolved rapidly over the past decade (Smith and Johnson, 2024). A useful textbook framing is offered by Doe and Roe (2020).

### 1.2 Thesis overview

Provide an overview of the three constituent papers and how they relate to one another. Per policy section 13.10, include here the rationale for submitting in journal format and an account of how the thesis has been constructed.

### 1.3 Thesis structure

Summarise the structure of the remaining chapters.

## References

Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.

Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.

Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.

Smith, J.A. and Johnson, M.B. (2024) 'An example journal article', *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

# Chapter 2

## Title of the first paper

### Abstract

*State the question, methods, principal findings, and contribution in 150–250 words. The abstract is rendered in italic and indented to match the conventional layout of a paper-format abstract.*

### 2.1 Introduction

Set up the research question. The literature on this topic has grown rapidly (Smith and Johnson, 2024). A textbook framing is provided by (Doe and Roe, 2020); the present paper extends this by [your contribution]. A complementary perspective is offered by (Park, 2022).

### 2.2 Methodology

The model is estimated by ordinary least squares. The estimating equation is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i \quad (2.1)$$

where  $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$  are independent and identically distributed error terms. Equation 2.1 is the workhorse specification.

Inline mathematics also work — for example, the t-statistic is  $t = \hat{\beta}/\text{SE}(\hat{\beta})$ .

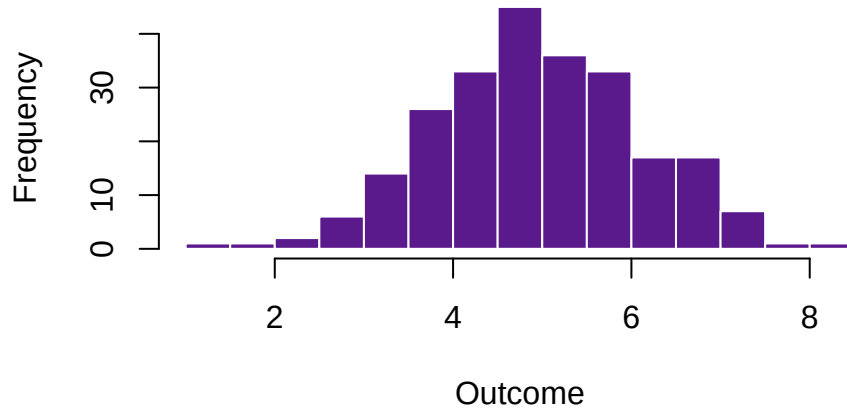


Figure 2.1: Distribution of the outcome variable.

## 2.3 Data

The sample comprises  $n = 240$  observations drawn from three sources. Descriptive statistics are reported in Table Table 2.1.

Table 2.1: Descriptive statistics of the sample.

| Variable       | Mean | SD  | Min | Max |
|----------------|------|-----|-----|-----|
| Age            | 27.3 | 3.1 | 22  | 39  |
| Programme year | 2.4  | 1.0 | 1   | 4   |
| Outcome score  | 4.8  | 1.2 | 1.5 | 6.9 |

## 2.4 Results

Figure Figure 2.1 shows the distribution of the outcome variable.

The estimated coefficients are reported in Table Table 2.2.

Table 2.2: Estimated regression coefficients from the specification in Equation Equation 2.1.

| Coefficient | Estimate | Std. error | t-value | p-value |
|-------------|----------|------------|---------|---------|
| Intercept   | 4.92     | 0.15       | 32.8    | < 0.001 |
| $x_1$       | 0.34     | 0.10       | 3.4     | 0.001   |
| $x_2$       | -0.18    | 0.10       | -1.8    | 0.073   |

The slope on  $x_1$  is positive and significant at the 1% level (Smith and Johnson, 2024).

## 2.5 Discussion

Interpret the results. Discuss what the findings imply for theory and practice (Garcia and Kumar, 2023).

## 2.6 Conclusions

Summarise the contribution of this paper.

## References

- Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.
- Garcia, E. and Kumar, R. (2023) 'An example conference paper', *Proceedings of the example conference*, pp. 44–52.
- Park, M.-J. (2022) 'An example book chapter', in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
- Smith, J.A. and Johnson, M.B. (2024) 'An example journal article', *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

## Appendices

### 2.A Sample survey instrument

Reproduce the survey instrument used to collect the primary data.

### 2.B Robustness checks

Document any robustness checks. Use sub-sub-sections for grouped material:

#### 2.B.1 Excluding outliers

Outlier exclusion does not change the headline result.

#### 2.B.2 Bootstrap standard errors

Bootstrap standard errors confirm the parametric inferences.

# Chapter 3

## Title of the second paper

### Abstract

*State the question, methods, principal findings, and contribution in 150–250 words.*

### 3.1 Introduction

### 3.2 Literature review

### 3.3 Methodology

### 3.4 Data

### 3.5 Results

### 3.6 Discussion

### 3.7 Conclusions

### References

Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.



- Garcia, E. and Kumar, R. (2023) 'An example conference paper', *Proceedings of the example conference*, pp. 44–52.
- Park, M.-J. (2022) 'An example book chapter', in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.
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## **Appendices**

### **3.A Supplementary material**

### **3.B Additional analysis**

# Chapter 4

## Title of the third paper

### Abstract

*State the question, methods, principal findings, and contribution in 150–250 words.*

### 4.1 Introduction

### 4.2 Literature review

### 4.3 Methodology

### 4.4 Data

### 4.5 Results

### 4.6 Conclusions

### References

Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.

Garcia, E. and Kumar, R. (2023) ‘An example conference paper’, *Proceedings of the example conference*, pp. 44–52.

Park, M.-J. (2022) ‘An example book chapter’, in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.

Smith, J.A. and Johnson, M.B. (2024) ‘An example journal article’, *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.

## **Appendices**

### **4.A Supplementary material**

# Chapter 5

## Conclusions

### 5.1 Implications and future research

Summarise the implications of the three papers and outline directions for future research.

### 5.2 Summary

Provide a brief summary of the overall contribution of the thesis.

## References

Doe, J. and Roe, R. (2020) *An example reference book*. 2nd edn. Manchester: Example Press.

Garcia, E. and Kumar, R. (2023) 'An example conference paper', *Proceedings of the example conference*, pp. 44–52.

Park, M.-J. (2022) 'An example book chapter', in A. Editor and B. Editor (eds) *Handbook of examples*. Example Press, pp. 101–128.

Smith, J.A. and Johnson, M.B. (2024) 'An example journal article', *Example Journal of Research*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/exj.2024.18.3>.